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Dataset-aware Utopia modality contribution for imbalanced multimodal learning

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ABSTRACT

Multimodal imbalance is a critical issue in multimodal learning research. Over recent years, numerous modulation strategies have been proposed, with a core focus on minimizing disparities in contributions across modalities. However, we observe that in many datasets, the contribution proportions of modalities are inherently unequal. Consequently, diagnosing multimodal imbalanced learning based on the criterion of equal contributions and optimizing models by minimizing modality contribution disparities often result in suboptimal performance. To address this issue, we propose the concept of "Utopia Contribution", which estimates the utopia contribution distribution of each modality based on dataset-specific characteristics. This distribution serves as the optimization objective for modulation strategies, facilitating the comprehensive exploitation of information from all modalities. Specifically, based on the principle of population risk, we estimate the utopia contribution distribution of modalities in the given dataset by analyzing the impact of each modality's presence or absence on model predictions. Additionally, to enhance the generalizability of our method, we further propose a model-agnostic approach based on mutual information to estimate the factual contribution distribution of each modality. During training, we employ KL divergence (Kullback–Leibler divergence) to align the factual contribution distribution with the utopia contribution distribution. Extensive experiments on three benchmark datasets – IEMOCAP, CMU-MOSEI, and AVE – demonstrate the rationality, reliability, and effectiveness of our method.

1. Introduction

Human perception is inherently multimodal, as our brains integrate information from various senses to construct a more holistic understanding of the world [1–3]. In recent years, significant advancements have been made in the field of deep multimodal learning, with numerous studies concentrating on diverse fusion strategies for integrating multiple modalities. Despite these efforts, the field also encounters many challenges. Studies [4–6] revealed that multimodal models, trained with a unified optimization objective across modalities, can sometimes perform worse than their unimodal counterparts. This paradox may arise when one modality is more informative and thus overshadows the learning of other modalities [7,8], potentially impeding the model's performance gains from multimodal information fusion.

To address this problem, empirical studies [9–12] have attempted to modulate multimodal model training to balance the learning from multiple modalities. These studies typically monitor indicators – such as weights, gradients, and accuracy – for each modality within the model. By comparing the magnitudes of an indicator across modalities, they can diagnose imbalanced learning, defining the modality that contributes the most as the *dominant modality*, while the others are considered *weaker modalities*. And subsequently they devise targeted modulation strategies. The existing strategies can be broadly grouped into two categories: (1) *Enhancing Learning on Weaker Modalities*: It includes employing knowledge distillation to individual modality branches using either unimodal models [13] or multimodal representations [9], as well as optimizing each unimodal branch separately through multi-task learning [10,11]. These methods can effectively improve the model's feature extraction capabilities for weaker modalities. (2) *Constraining the Dominant Modality*. When imbalanced learning occurs, the approach [14] limits the model's updates to dominant modality by reducing its gradient updates, while the approach [11] constrains the model by masking representation of the dominant modality. Both methods indirectly promote the learning of weak modalities

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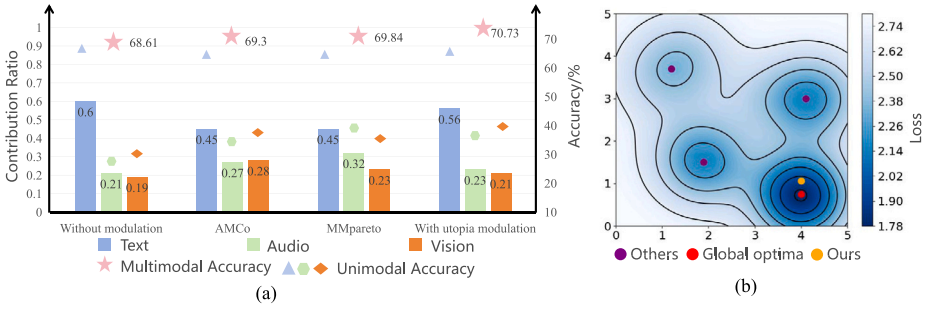


Fig. 1. (a) The contribution ratios of each modality and their corresponding accuracies in the MM-DFN model with three modulation strategies and without modulation strategy on the IEMOCAP dataset; (b) Schematic illustrations of optimization situations under different strategies.

by increasing the learning difficulty for the dominant one, thereby achieving a more balanced multimodal learning.

The key idea of aforementioned approaches is to minimize the disparities in contributions between modalities, yet, they overlook the fact that different modalities usually do not have an equal contribution to the prediction task in nature. Taking the emotion dataset IEMOCAP [15] as an example, which has three modalities, text is the dominant modality, while audio and vision are the weaker ones. We quantified the contribution ratios of each modality in the MM-DFN model [16] under various modulation strategies, along with their respective accuracies, as depicted in Fig. 1(a). It can be observed that the AMCo [11] and MMpereto [17] strategies, compared to no modulation, reduce the contribution differences between modalities and enhance the performance of the multimodal model. However, it is also noticeable that while the performance of the audio and visual modalities improves, the performance of the text modality declines, leading to suboptimal multimodal performance. We posit that the previous modulation strategies do not explicitly incorporate the optimal contribution ratios of each modality into the model optimization process, causing the models to fail in fully leveraging multimodal information and instead converging to local optima, as shown in Fig. 1(b). On the contrary, if the model was aware of the optimal contribution ratios during training, it could sidestep the pitfalls of local optima and approach, or even achieve, global optima.

Furthermore, the aforementioned methods also overlook the fact that the contributions of different modalities should be dataset-aware. For example, in the CMU-MOSEI dataset [18], vision contains richer emotional information than audio. In contrast, in the AVE dataset [19], audio features are more indicative of the underlying task than visual ones. This underscores the need for a dataset-aware approach to modality contribution estimation.

To address these issues, we introduce, for the first time, the concept of the Utopia Contribution Distribution (UCD) of modalities, which refers to the distribution of contributions from each modality to the model's prediction when it achieves global optimality. To obtain this UCD, we propose a novel method to estimate the utopia contribution ratio of each modality, incorporating a population risk metric that estimates the impact on the model's predictions of including or excluding the modality (see Appendix B for theoretical analysis). The obtained UCD is then employed as the optimization objective for existing modulation strategies, with the goal of enabling the model to fully leverage all modality information. Additionally, during training, it is imperative to estimate the Factual Contribution Distribution (FCD) of each modality to the model's current predictions. The optimization of the multimodal model thus becomes a process of aligning the FCD with the UCD. To this end, we introduce an estimation approach based on mutual information, which quantifies the factual contribution to predictions by gauging the proportion of information from the unimodal representation within the fused multimodal representation. An advantage of our proposal is its model-agnostic nature, which contrasts with other approaches that have limitations in terms of model applicability. The experiments conducted on three benchmark datasets for both classification and regression tasks demonstrate the rationality, reliability, and effectiveness of our proposed method.

The contributions of this paper are as follows:

- We, for the first time, introduce the concept of a multimodal Utopia Contribution Distribution (UCD) and formulate its mathematical model.
- We propose a dataset-aware method for estimating the UCD, providing an optimization objective for better balanced multimodal learning.
- We propose a model-agnostic method for estimating the Factual Contribution Distribution (FCD) based on mutual information estimation.

The organization of this paper is as follows. In Section 2, we review the related literature on multimodal learning and multimodal modulation strategies. Section 3 introduces the definition and the proposed method for estimating the UCD, as well as the approach for estimating the actual contribution distribution. In Section 4, we conduct extensive experiments to validate the reliability and efficacy of the proposed method. Section 5 discusses the limitations of our approach and potential solutions based on the experimental results. The paper concludes in Section 6 with a summary of its contributions and research significance.

2. Related works

2.1. Multimodal learning

Multimodal learning remains one of the essential challenges in artificial intelligence and machine learning, driving great research effort focused on fusing and leveraging information from multiple modalities [20,21]. This field has given rise to diverse research directions, addressing key tasks such as audio-visual scene understanding [22], visual question answering [23], and multimodal emotion recognition [18], to name a few. Researchers endeavor to model the relationships between modalities using transformer-based modules or graph-based fusion methods. However, existing joint training approaches often fail to fully leverage the information of all modalities, leading to suboptimal performance.

2.2. Imbalanced multimodal learning

The observed phenomenon has prompted researchers to investigate the underlying reasons. For instance, Huang et al. [2] examined the suboptimal performance through the lens of modality competition, while Peng et al. [14] attributed it to differing convergence rates driven by variance across modalities. To address the challenges of imbalanced multimodal learning, various approaches have been proposed, broadly grouped into two categories:

(1) *Enhancing the weaker modalities.* Several early efforts employed distillation techniques to bolster the learning efficacy of weaker modalities. For instance, Du et al. [13] introduced additional unimodal classifiers to refine the training of multimodal models, while Su et al. [9] leveraged features from multimodal teachers to guide the learning of weaker modalities. However, these methods incurred significant extra computational overhead. Alternative approaches, such as those by Fang et al. [10] and Zhou et al. [11], adopted multi-task learning to optimize unimodal performance independently, but were limited by the fusion strategies or model architectures. Fan et al. [24] proposed the use of prototypes to represent general class features, accelerating learning by refining weaker modality clustering toward its prototype. Wei et al. [17] proposed the MMpareto, integrating multimodal and unimodal objectives by constraining the direction and magnitude of gradients to balance the learning process. Fan et al. [25] explicitly enhanced weak modalities via a modal enhancement loss and prioritizes their participation in federated training.

(2) *Constraining the optimization speed of the dominant modalities.* Hu et al. [14] introduced OGM-GE, aiming to equalize the optimization of the model on all modalities by modulating the gradients of the dominant modality branches. However, this approach was limited to late fusion models. Wang et al. [26] proposed fine-grained adaptive gradient modulation, rebalancing modality gradients at the parameter level based on their dominance. Building on these advancements, Li et al. [27] defined the concept of mono-modal, providing deeper insights into the imbalance in multimodal learning. They subsequently employed Shapley values to estimate every modal contributions and constrained gradient updates of dominant modality accordingly. Huang et al. [28] proposed InfoReg, a method that regulates the Fisher Information of dominant modalities during the "prime learning window" to slow their information acquisition, thereby balancing multimodal learning and enhancing weak modalities. Wei et al. [29] softly reinitialized encoders of well-learned modalities to reduce over-reliance on them, balancing training emphasis across modalities.

There are also some works focus on these two perspectives simultaneously. Huang et al. [12] proposed a multi-level gradient calibration framework to optimize joint training in multi-modal multi-task learning. It balances task conflicts via inter-task gradient calibration and mitigates modality bias via intra-modality gradient calibration. Yang et al. [30] dynamically evaluated modality significance using mutual information rates and employ non-uniform adaptive sampling to update subnetworks, suppressing dominant modalities while enhancing non-dominant ones.

The aforementioned methods employed various means to diagnose whether a model exhibits imbalanced learning and subsequently apply modulation strategies to reduce the disparities in contributions among modalities. However, these approaches may overlook the inherent imbalance between modalities present in many datasets, and thus cannot explicitly define what constitutes an "optimal balanced state". In contrast to these indirect optimization objectives, we propose the concept of a Utopian Contribution Distribution, providing a direct optimization objective to guide modulation strategies effectively.

2.3. Factual modal contribution estimation

Accurately estimating the Factual Contribution Distribution (FCD) of modalities to a model's performance is essential for addressing multimodal imbalanced learning and optimizing information fusion strategies. Most existing methods are gradient-based, where factual contributions are estimated from the gradient value of each modality influencing on the model's output, such as OGM [14]. Higher gradient values generally indicate a greater contribution to the model's prediction. Other approaches quantify modal contributions by assigning weights to different modality features at the neuron level within deep networks (e.g., Wang et al. [26]). However, these methods are often constrained by specific model architecture or fusion strategies, limiting their generalizability.

The performance-based methods can also estimate the factual contribution of individual modalities. An emerging study [27] employs Shapley values from cooperative game theory, considering all possible combinations of modalities. This approach reframes the optimization problem in multimodal learning as transforming competitive interactions among modalities into cooperative ones, with Shapley values estimating each modality's competitive and cooperative roles throughout training. However, the computational complexity of this method scales exponentially with the number of modalities, hindering the real-time calculation of each modality's contribution during training. Notably, the FCD estimated in the above methods is primarily used to diagnose imbalanced learning in the model, with their optimization objective remaining focused on minimizing disparities in contributions among modalities.

Therefore, we propose an FCD estimation method grounded in mutual information [31], which is both computationally efficient and compatible with a variety of fusion frameworks. Unlike previous approaches that use FCD solely to diagnose imbalanced learning, our method aligns FCD with the UCD, serving as a direct optimization objective.

3. Methods

3.1. The problem formulation

Firstly, we formulate the multi-modal learning problem. Given a dataset $S = \{(x_i, y_i)\}_{i=1}^{k}$ comprising k samples, where x represents an input sample and y its corresponding label. Each x_i consists of m modalities, denoted as $x_i = \{x_i^1, x_i^2, \ldots, x_i^m\}$. Let $G = \{g : X \to Z\}$ represent the set of mapping functions from the input space to the latent space, and $H = \{h : Z \to Y\}$ denote the set of task mapping functions. We define $g^* \in G$ and $h^* \in H$ as the optimal mappings. Following the Empirical Risk Minimization (ERM) principle [32], the learning objective is to identify a pair of h and g that minimizes the empirical risk:

$$\min \hat{r} \triangleq \frac{1}{k} \sum_{i=1}^{k} l(h \circ g(x_i), y_i), \tag{1}$$

where l measures the gap between the model's predictions and the labels. $h \circ g = h(g(x))$ represents the combined operation of h and g.

Following previous studies, the FCD of all modalities to the model's prediction is denoted as $F = (F^1, \ldots, F^m)$, subject to $\sum_{j=1}^{m} F^j = 1$. Due to intrinsic modality disparities and model biases, it is common that $F^1 \neq \ldots \neq F^m$. Conventional multimodal modulation strategies aim to minimize the variance among $\{F^1, \ldots, F^m\}$, with the optimization objective formulated as:

$$\min \left[\hat{r}(h \circ g) + \gamma s^2(F)\right], \tag{2}$$

where $s^2(F)$ denotes the variance of F and γ is the coefficient. Notably, this optimization is inherently dataset-agnostic.

In practical scenarios, modality contributions are often unequal, and their distributions may vary significantly across datasets. We revisit the multimodal imbalance problem from a dataset-centric perspective, and define the *UCD* $I = (I^1, \ldots, I^m)$, subject to $\sum_{j=1}^{m} I^j = 1$, which represents the multimodal contribution distribution when the optimal model $h^* \circ g^*$ performs a prediction task on a given dataset. Since I is unknown during model training and often differs from F, the optimization (2) would not guide the model toward I. Consequently, the model may converge to a local optimum, hindering the effective utilization of all modalities. Clearly, steering F towards I during model training offers a more effective solution to the multimodal imbalance problem. Accordingly, we reformulate the optimization objective as:

$$\min \left[\hat{r}(h \circ g) + \varphi\left(\frac{1}{m} \sum_{j=1}^{m} d(F^j, I^j)\right)\right], \tag{3}$$

Algorithm 1 UCD estimation

Input: Dataset $S = \{(x_i^1, x_i^2, \dots, x_i^m), y_i\}_{i=1}^k$, unimodal encoder $\hat{g}_j (j = 1, \dots, m)$, multimodal classifier $\hat{h}_{ALL}$, unimodal classifier $\hat{h}_j$, missing modality classifiers $\hat{h}_{ALL-j}$.
Output: UCD $I = [I^1, I^2, \dots, I^m]$.
for $j = 1, 2, \dots, m$ **do**
  Train the unimodal encoder $\hat{g}_j$ and classifier $\hat{h}_j$ using $\{x_i^j\}_{i=1}^k$;
  Freeze the parameters of $\hat{g}_j$;
end for
Use encoders $\{\hat{g}_1, \dots, \hat{g}_m\}$ to train the multimodal classifier $\hat{h}_{ALL}$.
for $j = 1, 2, \dots, m$ **do**
  Use encoders $\{\{\hat{g}_1, \dots, \hat{g}_m\} - \hat{g}_j\}$ to train the classifier $\hat{h}_{ALL-j}$;
  Obtain $\Delta\mathfrak{r}(\hat{h} \circ \hat{g})$ using Eq. (5);
end for
Estimate UCD, $I = [I^1, I^2, \dots, I^m]$, using Eq. (6).

where $d(.)$ is a distance metric function and φ is the coefficient. Unlike the optimization (2), the reformulated objective (3) is dataset-aware, enabling the model to better leverage information across multiple modalities.

3.2. Utopia contribution estimations

To the best of our knowledge, there is currently no method capable of estimating the UCD I for a given dataset. Consequently, previous methods could not use I as an optimization objective during training. To address this challenge, we attempt to indirectly estimate I by evaluating the impact of each unimodal contribution on the model performance. Specifically, the utopia contribution of each modality is estimated by measuring the change in model prediction results when that modality is present or absent, please refer to Appendix B for the theoretical analysis.

To this end, we employ population risk [33] to evaluate the model performance under different inputs.

Given $\hat{r}(h \circ g)$, its corresponding population risk is defined as:

$$\mathfrak{r}(h \circ g) = \mathbb{E}_{(x_i, y_i)}[\hat{r}(h \circ g)]. \tag{4}$$

The change in population risk can be interpreted as the impact of a modality on the model's prediction in a multimodal environment, thereby reflecting its contribution to the prediction. For a dataset with m modalities, we train a model to minimize empirical risk using two types of inputs: the representation of complete modalities and the representation excluding modality j ($j = 1, \dots, m$). This results in two sets of encoders and classifiers: $(\hat{h}_{\text{ALL}}, \hat{g}_{\text{ALL}})$ and $(\hat{h}_{\text{ALL}-x^j}, \hat{g}_{\text{ALL}-x^j})$, respectively. The change in the model's population risk is then:

$$\Delta\mathfrak{r}(\text{ALL}, \text{ALL} - x^j) = \mathfrak{r}(\hat{h}_{\text{ALL}-x^j} \circ \hat{g}_{\text{ALL}-x^j}) - \mathfrak{r}(\hat{h}_{\text{ALL}} \circ \hat{g}_{\text{ALL}}). \tag{5}$$

The UCD of all modalities in the dataset can be obtained by normalizing the change:

$$I = \text{Normalize}([\Delta\mathfrak{r}(\text{ALL}, \text{ALL} - x^1), \dots, \Delta\mathfrak{r}(\text{ALL}, \text{ALL} - x^m)]). \tag{6}$$

The pseudocode of implementation is shown in the Algorithm 1, and the key steps are detailed in below:

Train the unimodal encoders $[\hat{g}_1, \dots, \hat{g}_m]$: Due to the variety and complexity of encoder architectures and the fusion strategies in existing multimodal models, it is challenging to isolate the unimodal encoders from the model. To simulate a fixed unimodal encoder $\hat{g}_j$, where $j \in 1, \dots, m$, only the representation of modality j is fed into the model and using zero-padding to represent the absence of other modalities. Concurrently, a unimodal classifier $\hat{h}_j$, composed of fully connected layers, is then attached to $\hat{g}_j$ for training.

Train the multimodal classifiers $[\hat{h}_{\text{ALL}-x^1}, \dots, \hat{h}_{\text{ALL}-x^m}, \hat{h}_{\text{ALL}}]$: With the unimodal encoders in place, we proceed to construct m sets of fused representations, each lacking one modality, as well as a set of complete-multimodal representations, by concatenating the outputs of $[\hat{g}_1, \dots, \hat{g}_m]$. For each set of representations, a corresponding multimodal classifier consisting of a single fully connected layer is trained. Specifically, we train m missing-modality classifiers $[\hat{h}_{\text{ALL}-x^1}, \dots, \hat{h}_{\text{ALL}-x^m}]$, along with a complete-modality classifier $\hat{h}_{\text{ALL}}$.

Estimate the UCD I: From the above steps, $m + 1$ pairs of $(\hat{h}, \hat{g})$ can be derived. According to Eq. (5), we can estimate the utopia contribution I^j of each modality j to the model's prediction. Finally, the obtained $\{I^1, \dots, I^m\}$ are normalized to derive the UCD, I.

3.3. Factual contribution estimation

During training, it is also essential to estimate the real-time factual contribution of each modality to quantify their deviation from the utopia contributions. To ensure compatibility across various frameworks using different fusion strategies, we employ the mutual information as a measure. Given a sample $x = \{x^1, x^2, \dots, x^m\}$ with m modalities, its multimodal fused representation z used for prediction, and their joint distribution $p(x^j, z)$ $(j = 1, \dots, m)$, the mutual information, $MI(x^j; z)$, represents the statistical dependence between x and z. In other words, $MI(x^j; z)$ quantifies how much information in z is related to x^j. A larger $MI(x^j; z)$ indicates a greater influence of the modality x^j on the prediction, making it a reliable indicator of the factual contribution of each modality.

To estimate it, we employ a score function to evaluate the correlation between the normalized fusion vector and input vector. This approach enables the estimation of a lower bound for mutual information.

$$\begin{aligned} MI(x^j, z) &= D_{KL}\left(p(x^j, z) \parallel p(x^j)p(z)\right) \\ &= \mathbb{E}_{p(x^j, z)}\left[\log \frac{p(x^j, z)}{p(x^j)p(z)}\right] \\ &\geq \log N + \mathbb{E}\left[\log \frac{u(x^j, z)}{\sum_S u(x^j, z)}\right] \\ &\triangleq I_{\text{InfoNCE}} \end{aligned} \tag{7}$$

where,

$$u(x^j, z) = \exp\left(\overline{z}(\overline{x^j})^T\right), \qquad \overline{x^j} = \frac{x^j}{\|x^j\|_2}, \quad \overline{z} = \frac{z}{\|z\|_2}. \tag{8}$$

The FCD, F, can be derived by normalizing all $MI(x^j, z)$,

$$F = (F^1, \dots, F^m) = \text{Normalize}\left[MI(x^1, z), \dots, MI(x^m, z)\right]. \tag{9}$$

3.4. Modulation strategy

As mentioned above, the goal of multimodal imbalanced learning is to align F with I. To achieve this, we minimize the KL divergence to reduce the distance between them,

$$L_{\text{KL}}(I, F) = \sum_{j=1}^{m} I^j \log\left(\frac{I^j}{F^j}\right). \tag{10}$$

An additional advantage of KL divergence is its adaptability to multimodal models with various fusion strategies.

For other methods addressing the imbalanced multimodal problem, their original balancing criteria can be simply replaced by the UCD, I, obtained in Section 3.2, after which their respective modulation strategies can be applied.

Finally, the joint training of the KL divergence loss, L_{KL}, and the task-specific loss, L_{task}, further enhances the effectiveness of the optimization objective, providing a more refined target for model training.

$$L_{\text{total}} = L_{\text{task}} + \alpha L_{\text{KL}}, \tag{11}$$

where α is a hyperparameter that controls the trade-off between two losses.

4. Experiments

4.1. Datasets

To evaluate the reliability and effectiveness of our proposed methods, we conduct experiments on three multimodal datasets encompassing both classification and regression tasks.

Classification task:

IEMOCAP [15] is a commonly used English corpus for emotion recognition. It includes 151 videos of two-way conversations and 7433 utterances of ten unique speakers, which have three modalities: text, audio, and vision. We follow the previous work and use six emotion categories: happiness, sadness, neutral, anger, excited, and frustration.

AVE [19] is a video dataset for audio-visual event localization, which covers 28 event classes and consists of 4143 ten-second videos with both auditory and visual tracks as well as frame-level annotations. All videos were collected from YouTube.

Regression task:

CMU-MOSEI [18] contains 22 856 annotated video segments (utterances) from 5000 videos, 1000 individuals and 250 different topics. Each segment is manually annotated with a sentiment value ranging from −3 to +3, indicating the polarity (by positive/negative) and relative strength (by absolute value) of expressed sentiment.

4.2. Experimental settings

For IEMOCAP and CMU-MOSEI datasets, librosa [34] is employed to extract the acoustic features. Librosa offers a suite of audio analysis tools for feature extraction, including MFCCs, Mel spectrograms, chroma features, spectral contrast, and pitch estimation. Specifically, we employ librosa to derive 64-dimensional acoustic features from each utterance. Textual features are extracted using the pre-trained DeBERTa [35] model, which builds upon BERT [36] and RoBERTa [37] with disentangled attention mechanisms and an enhanced mask decoder. Leveraging DeBERTa-large, we encode word sequences into 1024-dimensional textual features per utterance, capitalizing on its success in natural language understanding and generation. And the OpenFace [38] model is applied for visual features, which utilizes deep neural networks to perform real-time facial landmark detection, head pose estimation, facial action unit recognition, and gaze estimation. Subsequently, we aggregate frame-level facial features into 64-dimensional utterance-level features through element-wise averaging.

For AVE dataset, we employ a VGG-19 network [39], pre-trained on ImageNet, to extract features from the pool5 layer for each one-second visual segment, using 16 sampled RGB video frames. Then global average pooling is applied over these 16 frames to produce a single $512 * 7 * 7$ feature map. Temporal visual features are extracted by C3D [40], which is capable of learning spatio-temporal visual features. For each one-second audio segment, a 128-dimensional audio representation is extracted using a VGG-like network pre-trained on AudioSet.

The code was implemented in Pytorch 1.12, and experiments were run on a single NVIDIA 3090 GPU.

4.3. Baselines and evaluation metrics

Most existing modulation strategies are designed as plug-in forms, enabling combination with various models to evaluate the modulation efficacy. Therefore, we select multiple modulation strategies and combine them with various multimodal models for comprehensive analysis.

First, we choose diverse models, including early fusion, middle fusion, and late fusion models. For the early fusion model, DialogueRNN [41] is used for the IEMOCAP dataset, which simply concatenates modalities at the input end. For the CMU-MOSEI and AVE datasets, we constructed an early fusion model based on the Transformer module, the specific structure of which is detailed in Appendix. In terms of middle fusion models, CORECT [42] is used for both IEMOCAP and CMU-MOSEI, and MM-DFN [16] is used for the IEMOCAP dataset. As for the AVE dataset, the PSP model [43] is used, which incorporates sophisticated techniques for aligning audio and visual representations during fusion. We also constructed a Transformer-based late fusion model, designed to be compatible with all three datasets.

We select eight emerging modulation strategies, OGM [14], AMCo [11], AGM [27], FAGM [26], CRAMT [44], MMpareto [17], InfoReg [28] and DRBML [29]. Since OGM, AGM, FAGM, MMpareto, InfoReg and DRBML introduce criteria for measuring the multimodal imbalanced state, we replace their original criteria with the UCD, I, estimated by our method, followed by applying their respective modulation strategies for experimental validation. Instead, as CRAMT relies on a loss constraint rather than an explicit measurement criterion, we incorporate the KL divergence loss from Section 3.4 into CRAMT to align it with I.

Following previous studies, we employ Accuracy (ACC) and F1-score (F1) as our primary evaluation metric for classification task while mean absolute error (MAE), correlation (Corr) and R-squared (R^2) for regression task.

Table 1
Estimating the UCDs by different models on three datasets. The small variances indicate our UCD estimation method is robust and model-agnostic. And, the varied UCDs on different datasets demonstrate that the estimated UCD is dataset-aware.

Datasets	Model	I^T	I^A	I^V
IEMOCAP	DialogueRNN	0.54	0.23	0.23
	CORECT	0.53	0.26	0.21
	MM-DFN	0.55	0.24	0.21
	Late-fusion	0.56	0.24	0.20
	Variance	0.0002	0.0003	0.0002
CMU-MOSEI	Early-fusion	0.50	0.23	0.27
	CORECT	0.52	0.22	0.26
	Late-fusion	0.52	0.20	0.28
	Variance	0.0001	0.0002	0.0001
AVE	Early-fusion		0.56	0.44
	PSP		0.53	0.47
	Late-fusion		0.57	0.43
	Variance		0.0003	0.0003

4.4. Reliability analysis

4.4.1. Robustness of UCD estimation

To verify the robustness of the UCD estimation method proposed in Section 3.2, we conduct evaluation using at least three models on three datasets, respectively. The results are shown in Table 1. These models perform as the unimodal encoders and multimodal classifiers in Algorithm 1. Additionally, we calculated the variances of the utopia contributions derived from different models on each dataset. One can observe that all variance values are small, indicating that our UCD estimation method is robust and model-agnostic for a given dataset. This is highly valuable in practical applications, as it is not necessary to recalculate the UCD for different models. For a given dataset, the UCD can be derived from a simple model and subsequently applied to optimize other models.

4.4.2. Capability of dataset-aware

As mentioned earlier, our proposed UCD is dataset-aware. The results in Table 1 also confirm this. Specifically, on the CMU-MOSEI dataset, the visual modality plays a more important role in model predictions compared to the audio modality. Conversely, on the other two datasets, especially on the AVE dataset, the audio modality demonstrates a clear advantage over the visual modality. These observations confirm that the UCD derived by our method is indeed dataset-aware.

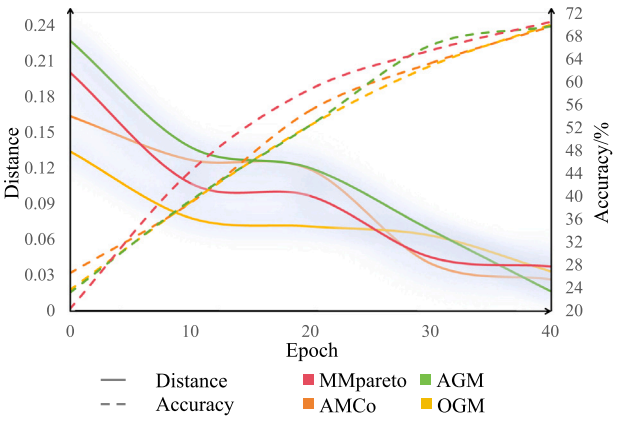


Fig. 2. The impact of multimodal contribution changes on model performance when applying UCD to four modulation strategies. The horizontal axis represents training iterations, the left vertical axis denotes the average distance between FCD and UCD, $\frac{1}{m}\sum_{i=1}^{m}|F^i - I^i|$, and the right vertical axis shows the model's accuracy. Solid lines represent changes in the average contribution distance during training, while dashed lines show the changes in accuracy. The orange, yellow, green, and red lines are corresponded to the base model with four modulation strategies: AMCo, OGM, AGM, and MMpareto, respectively.

4.4.3. Rationality of UCD

To validate the rationality of the UCD, we observe during the training whether it could assist in optimizing other modulation strategies. Thus, the optimization objectives of four recently proposed modulation strategies are replaced with UCD and the state-of-the-art mid-fusion model, MM-DFN [16], is used as the base model. We then evaluate how changes in modality contributions during training affect the model performance, as shown in Fig. 2. It can be observed that, as training progressed, the FCDs of the base model using different modulation strategies gradually approached the UCD. Correspondingly, the accuracy of these models steadily improved and eventually stabilized. These results demonstrate that UCD can be applied to various modulation strategies and effectively guide model training to optimize the contributions of each modality, thereby enhancing the overall performance of multimodal models.

4.5. Effectiveness of our proposed method

To evaluate the effectiveness of incorporating the UCD as the optimization objective on existing modulation strategies, we conduct comprehensive experiments combining various models and modulation strategies mentioned in Section 4.3 on the IEMOCAP, CMU-MOSEI, and AVE datasets. Tables 2–4 first present the estimated UCD for each dataset, which are the average of the UCDs derived by the models in Table 1. Next, the tables list the final FCDs, F, for different models when using the original settings of modulation strategies and when applying UCD as the optimization objective for these strategies, as well as the distance ΔDis between FCD and UCD. Additionally, we also report the results of the proposed optimization strategy, which directly minimizes the distance between FCD and UCD using KL divergence.

It can be observed that, compared to not using any modulation strategy, existing modulation strategies enhance multimodal learning performance. However, they tend to equalize the contributions of different modalities. Incorporating the UCD as an optimization objective further improves the performance of all modulation strategies, as indicated by the positive ΔAcc values. On the IEMOCAP dataset, incorporating UCD shows a more significant impact on early fusion models (e.g., DialogueRNN) and late fusion models. For mid-fusion models such as CORECT and MM-DFN, the distance between F and I, ΔDis, is already small when using the original modulation settings, thus the role of UCD is relatively minor. In contrast, on the CMU-MOSEI dataset, UCD is more pronounced in mid-fusion models. Specifically, the ΔDis is larger for the CORECT model when using original modulation settings, making UCD more meaningful for improving its performance. For the AVE dataset, which contains only two modalities with relatively balanced contributions, the effect of UCD on all three models is less significant.

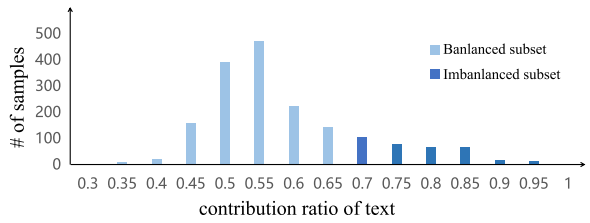


Fig. 3. The distribution of samples with diverse text-to-audio/visual contribution ratios in the IEMOCAP dataset.

We can also observe that mid-fusion models consistently achieve superior performance over early and late-fusion models. Our proposed KL divergence approach achieves the best results across all datasets.

4.6. Effectiveness of UCD on severely imbalanced multimodal data

The analysis in Section 3.2 indicates that for data where the contributions among modalities are relatively balanced, our proposed UCD strategy functions similarly to other modulation strategies by minimizing contribution disparities among modalities. However, for data with significant multimodal imbalance, incorporating UCD is more meaningful.

To investigate this further, we divided the test set of IEMOCAP into two subsets: a *balanced subset* and an *imbalanced subset*. The division is based on using MM-DFN as the base model with six modulation strategies, and calculating their FCDs. The average of FCDs is used as the selection criterion. Fig. 3 depicts the distribution of samples with diverse text-to-audio/visual contribution ratios, appearing in a right-skewed distribution. Thus, samples with a dominant modality contributing more than 70% are selected into the imbalanced subset, while others are selected into the balanced subset. We then evaluate the classification performance on these two subsets when UCD is applied.

As shown in Table 5, the role of UCD is minor on the balanced subset. In contrast, on the imbalanced subset, UCD improved the accuracy ranging from 0.56% to 2.05% across various strategies. This substantiates the necessity of UCD for effectively learning from imbalanced samples.

4.7. Case study

The above experiments statistically validate the effectiveness of UCD as an optimization objective. Here, we further visualize the changes in modality contributions for individual samples before and after applying UCD. We continue to use MM-DFN as the base model and apply three state-of-the-art modulation strategies.

Fig. 4 showcases some results, $\overline{F}$ represents the averaged FCDs in an utterance. It can be observed that when the model without a modulation strategy, the text modality significantly dominates the model's predictions. Incorporating modulation strategies tends to equalize the contributions across modalities. After employing UCD, the FCD falls between the two scenarios in above, not forcibly restricting the text modality information but enhancing the contribution of weaker modalities to the model prediction.

5. Discussion

Although our UCD more closely approximates the genuine distribution of modality contributions in the data than previous methods

Table 2
Performance comparisons of divers models and modulation strategies with and without UCD (T: 0.55; A: 0.24; V: 0.21) on IEMOCAP. ΔDis is the distance between F and I.

Model	Strategy	Using original settings						Using UCD							
		Acc↑	F1↑	F^T	F^A	F^V	ΔDis	Acc↑	ΔAcc	F1↑	$\Delta F1$	F^T	F^A	F^V	ΔDis
DialogueRNN	*No strategy*	*64.11*	*64.01*	*0.62*	*0.23*	*0.15*	*0.05*	–	–	–	–	–	–	–	–
	FAGM [26]	64.53	64.27	0.42	0.26	0.32	0.09	65.61	+1.08	64.54	+0.27	0.52	0.26	0.22	0.02
	AGM [27]	64.72	64.21	0.43	0.27	0.30	0.08	65.87	+1.15	64.67	+0.57	0.53	0.26	0.21	0.01
	+KL(ours)	–	–	–	–	–	–	66.12	–	65.91	–	0.54	0.24	0.22	0.01
CORECT	*No strategy*	*67.14*	*66.35*	*0.58*	*0.28*	*0.14*	*0.07*	–	–	–	–	–	–	–	–
	OGM [14]	67.59	66.90	0.45	0.26	0.29	0.06	68.01	+0.42	67.23	+0.33	0.50	0.26	0.24	0.02
	FAGM [26]	67.81	66.98	0.42	0.25	0.33	0.08	68.36	+0.55	67.16	+0.18	0.48	0.25	0.27	0.05
	AGM [27]	68.02	67.31	0.40	0.29	0.31	0.10	68.64	+0.62	67.43	+0.12	0.49	0.24	0.27	0.04
	AMCo [11]	68.17	67.02	0.46	0.26	0.28	0.06	68.74	+0.72	67.27	+0.25	0.49	0.25	0.26	0.04
	CRMT [44]	68.39	67.71	0.43	0.26	0.31	0.08	68.70	+0.31	68.26	+0.55	0.51	0.24	0.25	0.03
	MMPareto [17]	68.87	67.98	0.42	0.28	0.30	0.08	69.05	+0.18	68.67	+0.69	0.49	0.28	0.23	0.04
	InfoReg [28]	68.01	67.13	0.43	0.31	0.26	0.08	68.39	+0.38	67.34	+0.21	0.50	0.25	0.25	0.03
	DRBML [29]	67.92	67.03	0.43	0.35	0.22	0.08	68.73	+0.81	67.25	+0.22	0.52	0.23	0.25	0.03
	+KL(ours)	–	–	–	–	–	–	69.16	–	68.94	–	0.53	0.19	0.28	0.01
MM-DFN	*No strategy*	*68.61*	*67.36*	*0.60*	*0.21*	*0.19*	*0.03*	–	–	–	–	–	–		
	OGM [14]	68.87	68.01	0.42	0.28	0.30	0.05	69.07	+0.20	68.19	+0.18	0.57	0.23	0.20	0.01
	FAGM [26]	69.04	68.25	0.41	0.22	0.37	0.07	69.49	+0.45	68.56	+0.31	0.52	0.25	0.23	0.02
	AGM [27]	69.24	68.38	0.43	0.21	0.36	0.07	69.86	+0.62	69.10	+0.72	0.53	0.22	0.25	0.02
	AMCO [11]	69.30	68.47	0.45	0.27	0.28	0.06	69.54	+0.24	69.40	+0.93	0.50	0.26	0.24	0.03
	CRMT [44]	69.67	69.10	0.43	0.29	0.28	0.06	70.10	+0.43	69.36	+0.26	0.52	0.27	0.21	0.03
	MMPareto [17]	69.84	69.36	0.45	0.32	0.23	0.04	70.35	+0.59	70.05	+0.69	0.52	0.25	0.23	0.02
	InfoReg [28]	69.36	68.32	0.41	0.33	0.26	0.09	70.16	+0.80	68.36	+0.04	0.54	0.23	0.23	0.01
	DRBML [29]	70.06	69.36	0.45	0.32	0.23	0.07	70.53	+0.47	69.43	+0.07	0.55	0.21	0.24	0.02
	+KL(ours)	–	–	–	–	–	–	70.73	–	70.21	–	0.56	0.23	0.21	0.01
Late-fusion	*No strategy*	*54.12*	*53.69*	*0.60*	*0.23*	*0.17*	*0.03*	–	–	–	–	–	–		
	OGM [14]	54.32	54.02	0.38	0.31	0.31	0.11	54.64	+0.32	54.14	+0.12	0.50	0.27	0.23	0.03
	FAGM [26]	54.69	54.17	0.42	0.29	0.29	0.08	55.52	+0.83	55.10	+0.93	0.52	0.26	0.22	0.02
	AGM [27]	55.13	54.91	0.41	0.29	0.30	0.08	56.14	+1.01	55.04	+0.13	0.55	0.25	0.20	0.01
	AMCo [11]	55.34	54.26	0.42	0.30	0.28	0.07	56.24	+0.90	56.03	+1.77	0.51	0.26	0.23	0.03
	CRMT [44]	56.68	56.08	0.40	0.29	0.31	0.10	56.20	+0.52	57.13	+1.05	0.52	0.26	0.22	0.02
	MMPareto [17]	56.78	56.34	0.43	0.24	0.33	0.08	57.01	+0.23	57.86	+1.52	0.53	0.27	0.20	0.02
	InfoReg [28]	54.90	54.11	0.47	0.21	0.23	0.04	55.20	+0.30	54.26	+0.15	0.52	0.26	0.22	0.02
	DRBML [29]	56.30	55.37	0.40	0.33	0.24	0.09	57.10	+0.80	55.67	+0.30	0.55	0.24	0.21	0
	+KL(ours)	–	–	–	–	–	–	57.31	–	58.20	–	0.53	0.25	0.22	0.01

Table 3
Performance comparisons on CMU-MOSEI (UCD, T: 0.52; A: 0.23; V: 0.25).

Model	Strategy	Using original settings							Using UCD								
		MAE↓	Corr↑	R^2 ↑	F^T	F^A	F^V	ΔDis	MAE↓	ΔMAE	Corr↑	$\Delta Corr$	R^2 ↑	ΔR^2	F^T	F^A	F^V	ΔDis
Early-fusion	*No strategy*	*0.5540*	*0.7014*	*0.5632*	*0.58*	*0.18*	*0.24*	*0.05*	–	–	–	–	–	–	–	–	–	–
	FAGM [26]	0.5531	0.7112	0.5668	0.45	0.25	0.30	0.04	0.5426	–0.0105	0.7137	+0.0025	0.5769	+0.0101	0.50	0.21	0.29	0.03
	AGM [27]	0.5513	0.7033	0.5713	0.49	0.23	0.28	0.05	0.5489	–0.0117	0.7150	+0.0136	0.5816	+0.0103	0.52	0.23	0.25	0.01
	+KL(ours)	–	–	–	–	–	–	–	0.5421	–	0.7121	–	0.5961	–	0.52	0.23	0.25	0.01
CORECT	*No strategy*	*0.5246*	*0.7939*	0.6016	*0.48*	*0.23*	*0.29*	*0.04*	–	–	–	–	–	–	–	–	–	–
	OGM [14]	0.5364	0.7931	0.6024	0.43	0.20	0.37	0.09	0.5102	–0.0082	0.8001	+0.0070	0.6112	+0.0088	0.46	0.24	0.30	0.04
	FAGM [26]	0.5330	0.7918	0.6064	0.42	0.24	0.34	0.07	0.5125	–0.0205	0.8013	+0.0095	0.6116	+0.0052	0.49	0.26	0.25	0.02
	AMCo [11]	0.5310	0.7913	0.6216	0.44	0.30	0.26	0.05	0.5092	–0.0218	0.8005	+0.0092	0.6253	+0.0037	0.48	0.25	0.27	0.02
	AGM [27]	0.5283	0.7991	0.6117	0.43	0.34	0.23	0.06	0.5083	–0.0200	0.8011	+0.0020	0.6221	+0.0104	0.49	0.22	0.29	0.04
	MMPareto [17]	0.5183	0.7946	0.6345	0.44	0.23	0.33	0.07	0.5002	–0.0181	0.8022	+0.0076	0.6386	+0.0041	0.47	0.25	0.28	0.03
	InfoReg [28]	0.5230	0.7913	0.6201	0.44	0.20	0.36	0.07	0.5101	–0.0129	0.7996	+0.0083	0.6259	+0.0058	0.51	0.22	0.27	0.01
	DRBML [29]	0.5112	0.7962	0.6193	0.45	0.22	0.33	0.05	0.5036	–0.0076	0.8016	+0.0054	0.6203	+0.0010	0.52	0.23	0.25	0
	+KL(ours)	–	–	–	–	–	–	–	0.4951	–	0.8075	–	0.6412	–	0.48	0.25	0.27	0.02
Late-fusion	*No strategy*	*0.5503*	*0.7733*	*0.5663*	*0.54*	*0.19*	*0.27*	*0.04*	–	–	–	–	–	–	–	–	–	–
	OGM [14]	0.5446	0.7763	0.5736	0.44	0.24	0.32	0.06	0.5431	–0.0015	0.7781	+0.0018	0.5776	+0.0040	0.46	0.29	0.25	0.04
	FAGM [26]	0.5436	0.7762	0.5836	0.47	0.25	0.28	0.03	0.5417	–0.0019	0.7803	+0.0041	0.5862	+0.0026	0.55	0.24	0.21	0.02
	AMCo [11]	0.5421	0.7793	0.5897	0.46	0.20	0.34	0.07	0.5391	–0.0030	0.7820	+0.0027	0.5961	+0.0064	0.52	0.28	0.20	0.02
	AGM [27]	0.5404	0.7753	0.5869	0.42	0.25	0.33	0.06	0.5342	–0.0062	0.7802	+0.0049	0.5934	+0.0065	0.51	0.23	0.26	0.02
	MMPareto [17]	0.5331	0.7745	0.5931	0.46	0.23	0.31	0.05	0.5318	–0.0013	0.7803	+0.0041	0.6003	+0.0072	0.55	0.24	0.21	0.02
	InfoReg [28]	0.5460	0.7803	0.5867	0.42	0.26	0.32	0.07	0.5215	–0.0245	0.7904	+0.0101	0.5912	+0.0045	0.50	0.23	0.27	0.01
	DRBML [29]	0.5365	0.7836	0.5903	0.44	0.24	0.32	0.05	0.5337	–0.0028	0.7867	+0.0031	0.5968	+0.0065	0.55	0.22	0.23	0.02
	+KL(ours)	–	–	–	–	–	–	–	0.5313	–	0.7875	–	0.6027	–	0.51	0.24	0.25	0.01

that equalize contributions, our current method's focus on optimizing modality contribution distributions at the dataset level, while insufficiently addressing sample-level heterogeneity in modality contributions. As shown in Fig. 3, although the text modality dominates at the overall dataset level, its contribution decreases significantly or even becomes weak in certain samples. This sample-level fluctuation in modality contributions may lead to suboptimal results when using global statistics-based optimization strategies — applying dataset-level

Table 4
Performance comparisons on AVE (UCD, A: 0.55; V: 0.45).

Model	Strategy	Using original settings					Using UCD						
		Acc↑	F1 ↑	F^A	F^V	ΔDis	Acc↑	ΔAcc	F1 ↑	$\Delta F1$	F^A	F^V	ΔDis
Early-fusion	*No strategy*	*67.51*	*67.10*	*0.47*	*0.53*	*0.08*	–	–	–	–	–	–	–
	FAGM [26]	67.63	67.13	0.49	0.51	0.06	68.31	+0.68	67.32	+0.19	0.52	0.48	0.03
	AGM [27]	67.66	67.26	0.50	0.50	0.05	67.92	+0.26	67.43	+0.17	0.54	0.46	0.01
	+KL(ours)	–	–	–	–	–	68.78	–	68.06	–	0.55	0.45	0.00
PSP	*No strategy*	*74.21*	*73.36*	*0.46*	*0.54*	*0.05*	–	–	–	–	–	–	–
	OGM [14]	74.23	73.53	0.49	0.51	0.06	75.04	+0.77	74.41	+0.88	0.51	0.49	0.04
	FAGM [26]	74.31	73.97	0.48	0.52	0.07	75.21	+0.9	74.15	+0.18	0.52	0.48	0.03
	AMCo [11]	74.33	74.46	0.48	0.52	0.07	75.11	+0.78	75.13	+0.67	0.52	0.48	0.03
	AGM [27]	74.42	74.16	0.47	0.53	0.08	75.15	+0.73	75.04	+0.88	0.50	0.50	0.05
	CRMT [44]	74.64	74.35	0.48	0.52	0.07	75.01	+0.37	74.98	+0.63	0.57	0.43	0.02
	MMPareto [17]	74.65	74.17	0.47	0.53	0.08	75.11	+0.46	74.36	+0.19	0.56	0.44	0.01
	InfoReg [28]	74.03	73.03	0.50	0.50	0.03	75.10	+1.07	73.31	+0.28	0.53	0.47	0.01
	DRBML [29]	74.72	73.14	0.53	0.47	0.01	75.16	+0.44	73.28	+0.14	0.55	0.45	0
	+KL(ours)	–	–	–	–	–	75.39	–	74.69	–	0.54	0.46	0.01
Late-fusion	*No strategy*	*68.02*	*67.29*	*0.49*	*0.51*	*0.06*	–	–	–	–	–	–	–
	OGM [14]	68.38	66.36	0.51	0.49	0.04	69.12	+0.74	66.99	+0.63	0.52	0.48	0.03
	FAGM [26]	68.62	66.72	0.51	0.49	0.04	69.31	+0.69	67.03	+0.31	0.52	0.48	0.03
	AGM [27]	68.71	66.53	0.52	0.48	0.03	69.73	+1.02	67.23	+0.70	0.54	0.46	0.01
	AMCo [27]	68.83	67.15	0.51	0.49	0.04	69.35	+0.52	69.01	+1.86	0.52	0.48	0.03
	CRMT [44]	68.97	67.03	0.53	0.47	0.02	69.60	+0.73	68.53	+1.50	0.56	0.44	0.01
	MMPareto [17]	69.10	67.48	0.50	0.50	0.05	69.42	+0.32	69.03	+1.55	0.53	0.47	0.02
	InfoReg [28]	69.36	69.15	0.51	0.49	0.03	70.03	+0.67	69.24	+0.09	0.55	0.45	0
	DRBML [29]	69.10	69.21	0.51	0.49	0.03	69.91	+0.81	69.43	+0.22	0.52	0.48	0.02
	+KL(ours)	–	–	–	–	–	70.15	–	69.36	–	0.54	0.46	0.01

Table 5
Evaluation of UCD on imbalanced and balanced subsets. The base model is MM-DFN with eight modulation strategies.

	Balanced subset								Imbalanced subset							
Strategy	Without UCD			With UCD					Without UCD			With UCD				
	Acc↑	F1↑	F	Acc↑	ΔAcc	F1↑	$\Delta F1$	F	Acc↑	F1↑	F	Acc↑	ΔAcc	F1↑	$\Delta F1$	F
No strategy	66.13	65.70	F^T:0.51 F^A:0.29 F^V:0.20	–	–	–	–	– – –	68.93	68.71	F^T:0.62 F^A:0.22 F^V:0.16	–			– –	– – –
OGM [14]	68.45	68.23	F^T:0.40 F^A:0.30 F^V:0.30	68.60	+0.15	68.32	+0.09	F^T:0.46 F^A:0.28 F^V:0.26	68.19	68.12	F^T:0.43 F^A:0.23 F^V:0.27	68.75	+0.56	68.55	+0.43	F^T:0.53 F^A:0.28 F^V:0.19
FAGM [26]	69.21	68.74	F^T:0.43 F^A:0.26 F^V:0.31	69.63	+0.42	68.85	+0.11	F^T:0.46 F^A:0.30 F^V:0.24	68.32	68.41	F^T:0.45 F^A:0.27 F^V:0.28	69.78	+1.46	68.86	+0.45	F^T:0.54 F^A:0.24 F^V:0.22
AGM [27]	68.70	68.81	F^T:0.41 F^A:0.27 F^V:0.32	68.74	+0.04	69.02	+0.21	F^T:0.47 F^A:0.29 F^V:0.24	69.10	68.62	F^T:0.46 F^A:0.29 F^V:0.25	71.15	+2.05	69.17	+0.55	F^T:0.52 F^A:0.30 F^V:0.18
AMCo [11]	69.65	68.39	F^T:0.39 F^A:0.29 F^V:0.32	69.74	+0.09	68.68	+0.29	F^T:0.43 F^A:0.32 F^V:0.25	69.12	68.33	F^T:0.47 F^A:0.27 F^V:0.26	70.01	+0.89	68.99	+0.66	F^T:0.51 F^A:0.28 F^V:0.21
CRMT [44]	69.76	69.01	F^T:0.40 F^A:0.33 F^V:0.27	70.11	+0.35	69.09	+0.08	F^T:0.48 F^A:0.28 F^V:0.24	69.86	68.89	F^T:0.45 F^A:0.29 F^V:0.26	70.44	+0.58	69.25	+0.36	F^T:0.55 F^A:0.24 F^V:0.21
MMPareto [17]	70.03	69.11	F^T:0.43 F^A:0.27 F^V:0.30	70.14	+0.11	69.20	+0.09	F^T:0.45 F^A:0.30 F^V:0.25	69.26	68.56	F^T:0.45 F^A:0.26 F^V:0.24	71.23	+1.97	69.14	+0.58	F^T:0.54 F^A:0.26 F^V:0.20
InfoReg [28]	69.57	68.32	F^T:0.41 F^A:0.36 F^V:0.23	69.94	+0.37	68.42	+0.10	F^T:0.42 F^A:0.32 F^V:0.26	69.25	68.12	F^T:0.45 F^A:0.33 F^V:0.22	70.11	+0.89	69.13	+1.01	F^T:0.54 F^A:0.24 F^V:0.22
DRRML [29]	69.92	69.42	F^T:0.42 F^A:0.33 F^V:0.25	70.41	+0.49	69.55	+0.08	F^T:0.45 F^A:0.28 F^V:0.27	69.47	68.94	F^T:0.46 F^A:0.30 F^V:0.24	70.23	+0.76	69.41	+0.47	F^T:0.51 F^A:0.26 F^V:0.23

constraints to dominant modalities may inadvertently suppress the model's feature extraction capability for samples where these modalities are weaker. A potential solution is to stratify the dataset into distinct levels according to modality contribution distributions of different samples and implement tailored modulation strategies for each subset. We will explore this approach in future work.

6. Conclusion

In this paper, we address the challenge of multimodal imbalanced learning by proposing a novel dataset-aware estimation method for the utopia contribution distribution (UCD) of each modality in multimodal data, grounded in the principle of population risk, in which the UCD is derived from measuring the impact of each modality on model performance. This method goes beyond the limitations of previous works that suppress the learning of dominant modality information by minimizing the disparities in contributions across modalities, instead uses the UCD of multimodal data as the optimization objective for modulation strategies, thereby enabling a more holistic exploitation of multimodal information. Extensive experiments on diverse benchmark datasets and tasks substantiate the rationality, effectiveness, and

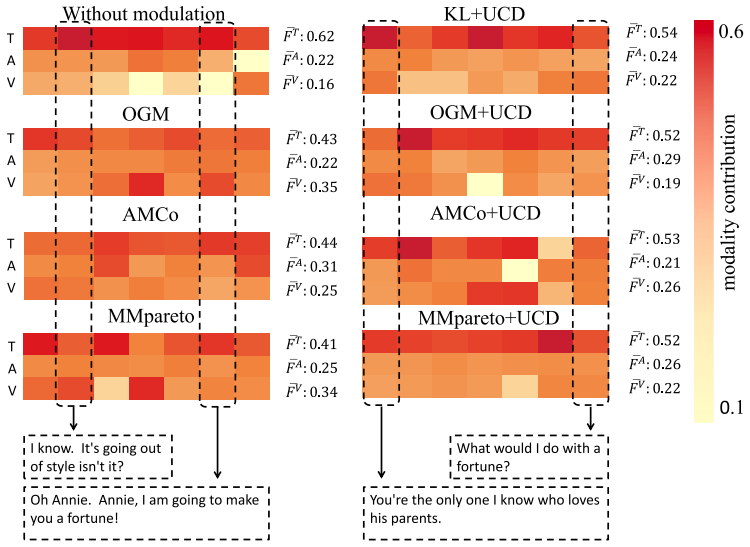


Fig. 4. Visualization of modality contributions within utterances from a dialogue in the IEMOCAP dataset when modulation strategies are with and without UCD.

robustness of the UCD. Notably, the necessity of UCD is highlighted on severely imbalanced data.

CRediT authorship contribution statement

Ying Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xuefeng Liang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yue Xu:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Xiuyun Lin:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xuefeng Liang reports financial support was provided by The research funding of GuangZhou DaYou Network Technology Co., Ltd, China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Framework

In order to comprehensively validate our proposal and existing modulation strategies, we constructed our own early fusion model and late fusion model based on transformers, as shown in Fig. A.5.

Appendix B. Theoretical analysis of the estimation of UCD

B.1. Definitions and notations

Optimal Mappings: Let $G = \{g : X \to Z\}$ represent the set of mapping functions from the input space to the latent space, and $H = \{h : Z \to Y\}$ denote the set of task mapping functions. g^* and h^* denote the optimal mappings of the multimodal model. In this context, the distribution of contributions from each modality to the model's prediction can be regarded as the UCD.

Trained Mappings: Denotes $\hat{g}_{\text{ALL}}$ and $\hat{h}_{\text{ALL}}$ as the mappings for complete-modality representations, and $\hat{g}_{\text{ALL}-j}$ and $\hat{h}_{\text{ALL}-j}$ as the mappings for single-modality missing representations, where j is the index of the missing modality, $j = 1, \ldots, m$.

Risk Function: $\mathfrak{r}(h \circ g)$ is the population risk that can be used to evaluate the performance of the mapping $h \circ g$.

B.2. Theoretical analysis

The model optimization is label-driven, aiming to obtain a mapping $h \circ g$ that is as close as possible to the optimal mapping $g^* \circ h^*$. It should be noted that while the utopia mappings, g^* and h^*, cannot be directly obtained during the model learning process, $g^* \circ h^*$ can serve as a reference point for model performance analysis. Therefore, we define d as the distance between $h \circ g$ and the optimal mapping $g^* \circ h^*$, serving as an indicator of the quality of the learned representations.

$$d(h \circ g, h^* \circ g^*) = \inf_{h \in H, g \in G} \left[\mathfrak{r}(h \circ g) - \mathfrak{r}(h^* \circ g^*) \right], \tag{B.1}$$

where inf denotes the lower bound. It indicates that the smaller the distance d, the higher the quality of the learned representations, and consequently, the lower the population risk of the model.

Therefore, by comparing the deviation between the mappings of complete multimodal representations and those with a single missing modality against the optimal mapping, $g^* \circ h^*$, in the feature space, we can quantitatively evaluate the impact of missing modalities on the model's predictive performance.

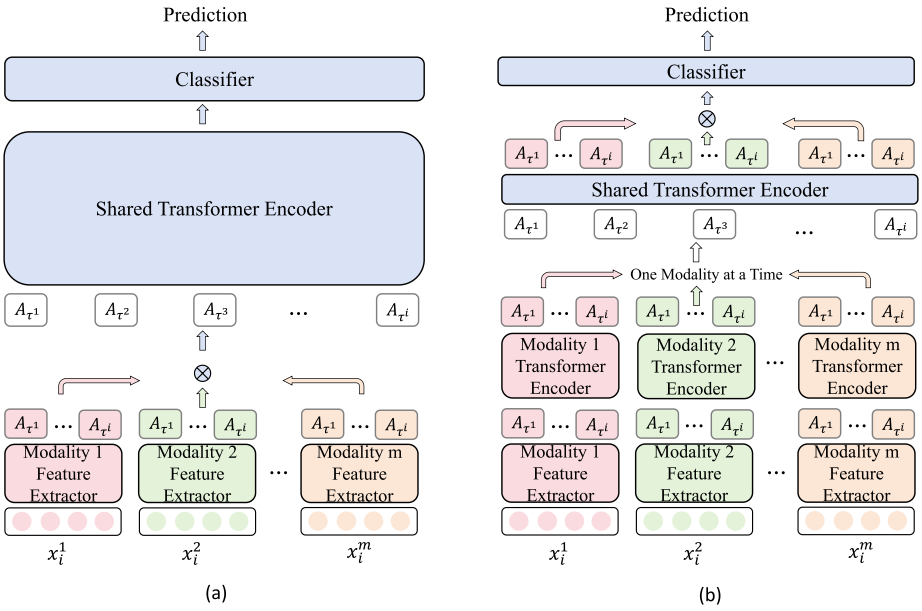


Fig. A.5. (a) Early fusion and (b) Late fusion models utilized for validation.

Specifically, the distance between the complete-modality representations and the optimal mappings is

$$d_{\mathrm{ALL}} = \inf_{\hat{h}_{\mathrm{ALL}} \in H, \hat{g}_{\mathrm{ALL}} \in G} \left[\mathfrak{r}(\hat{h}_{\mathrm{ALL}} \circ \hat{g}_{\mathrm{ALL}}) - \mathfrak{r}(h^* \circ g^*) \right], \tag{B.2}$$

and, the distance between single-modality missing representations and the optimal mappings is

$$d_{\mathrm{ALL}-x^j} = \inf_{\hat{h}_{\mathrm{ALL}-x^j} \in H, \hat{g}_{\mathrm{ALL}-x^j} \in G} \left[\mathfrak{r}(\hat{h}_{\mathrm{ALL}-x^j} \circ \hat{g}_{\mathrm{ALL}-x^j}) - \mathfrak{r}(h^* \circ g^*) \right]. \tag{B.3}$$

The deviation between the mappings of complete multimodal representations and those with a single missing modality against the optimal mapping is

$$\Delta d_j = d_{\mathrm{ALL}} - d_{\mathrm{ALL}-x^j}, \tag{B.4}$$

which is equivalent to the contribution of modality j to multimodal prediction. By canceling out the common factors, we then have

$$\Delta \mathfrak{r}_j = \mathfrak{r}(\hat{h}_{\mathrm{ALL}-x^j} \circ \hat{g}_{\mathrm{ALL}-x^j}) - \mathfrak{r}(\hat{h}_{\mathrm{ALL}} \circ \hat{g}_{\mathrm{ALL}}). \tag{B.5}$$

Data availability

Data will be made available on request.

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